CSE475: Machine Learning (Section-1) [Fall 26]

Project Title : (Task 1)

Graph-Enhanced Attention Network for OCT Classification

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Related Work Table

Title	Authors	Year	Dataset	Application area	Method/Model	Metrics	Strengths	Limitations	Research gap	Relation to your work
Classifying retinal diseases via pyramid vision graph convolutional network for optical coherence tomography images	Jin Qian, Lei Tao, Changhao Gong, Jun Xu, Yuemei Luo	2025	BOE/Srinivasan	Image-level OCT classification (3- and 4-class); no segmentation/localization	ViG-style pyramid GNN (PVGCN); Stem → feature-map nodes; kNN graph (K=9); max-relative graph conv; FFN vs over-smoothing; 4-stage pyramid; no attention; edges not anatomical	Own 10% test, averaging unstated. BOE Acc/P/R/F1 0.9954/0.9918/0.9877/0.9896. CELL 0.9787/0.9650/0.9555/0.9602. CELL baselines: VGG-16 0.9735, ResNet-18 0.9694, Graph Transformer 0.9592. Balanced training: Acc 0.9802, F1 0.9653. 5% training data: 0.9599	Ablations (GCN vs ViG, pyramid, K); data-fraction and balancing experiments; compact (11.47 M vs 138.36 M params for VGG-16); two datasets	Random image-level split; possible patient leakage (BOE: 45 subjects); numeric inconsistencies (counts, parameters); no repeated runs/CI; no interpretability or cross-dataset test; per-class results only as confusion matrix; graph overhead at inference; MRI/CT untested; expert comparison doesn't confirm same test images	No attention or CNN+attention comparison under same protocol; no patient-level evaluation; no per-class P/R tables; edge meaning unclear anatomically; no cross-dataset testing	Nearest OCT GNN baseline. Adopt: feature-map nodes, kNN with K sensitivity, FFN, balanced training. Yours adds attention + leakage-audited split. To show: graph helps beyond CNN(+attention) under identical conditions
GABNet: global attention block for retinal OCT disease classification	Xuan Huang, Zhuang Ai, Hui Wang, Chongyang She, Jing Feng, Qihao Wei, Baohai Hao, Yong Tao, Yaping Lu, Fanxin Zeng	2023	Kermany/UCSD	4-class OCT classification with external-device test; secondary DR grading (fundus)	CNN + attention. GAB: height/width/channel attention from avg/max pooling + depthwise-separable convs. GABNet = DSC residual blocks + GAB. GAB also plugged into Xception, EfficientNetV2B3; compared with SE, CBAM, CoorDatt	Weighted-average Acc/Recall/Spec/F1/AUC (weighted recall = accuracy). Complete data, no transfer: GABNet Acc 0.965, Spec 0.9883, AUC 0.9969, 9.361 M params; Xception 0.95; InceptionV3 0.96; ResNet50 0.941; EffNetV2B3 0.928. Transfer: Xception+GAB 0.99; EffNetV2B3+GAB 0.978. Xception, limited data: none 0.976, SE 0.978, CBAM 0.978, CoorDatt 0.985, GAB 0.99. External: GABNet 0.8047; Xception+GAB 0.9141	Controlled attention comparison (2 backbones, 2 data regimes); params and test time reported; external-device test; second task; Grad-CAM shown	Longest test time (7.26 s vs ~4.8 s); below transfer-learning models; external drop (Spectralis vs Cirrus); small gains on complete data (+0.3 to +1.0 pts); CBAM lowered Xception (0.977 → 0.961); no per-class metrics or repeated runs; patient handling not described; small external set (20 drusen); Grad-CAM qualitative only	No relational/graph module (attention only re-weights features); no lesion- or layer-level attention evaluation; no per-class recall for Drusen; no patient-level split description	Reference for attention component, weighted vs macro metrics, external-device testing. Yours (CNN+GCN+Attention) must show attention still helps with graph present; report macro/per-class metrics
Deep attention for enhanced OCT image analysis in clinical retinal diagnosis	Fatma M. Talaat, Ahmed Ali Ahmed Ali, Raghda ElGendy, Mohamed A. ELShafie	2025	OCT Retinal Disease Dataset,	4-class OCT classification; clinical diagnosis support	Attention-Based DenseNet (OCTNet) + L2 regularization; augmentation (rotation 20, shifts/shear/zoom 0.1, flip); batch 32, 50 epochs. Attention type and placement unspecified; algorithms illegible in file	With L2: test acc 0.9583, validation acc 0.9792 (0.9167 without). No precision/recall/F1/AUC/per-class results. Table 4 lists generic model-family accuracies without sources (CNN 97%, RNN 96%, LSTM 96.5%, Transformers 96.7%, Attention 95.95%, VAE	Kermany 4-class data; augmentation/training settings stated; limitations list (diversity, compute, real-time use, interpretability, benchmarks)	Not reproducible (no split sizes, attention definition, code); accuracy only; ablation described without numbers; 'clinical validation' claimed, none shown; Table 4 'OCT 98.7%' conflicts with own 0.9167-0.9792; unrelated passages (sleep health, autism,	Attention construction unspecified; no lesion focus check; no graph/relational modeling; no anatomical/layer analysis; no patient-le	Cautionary example. Specify attention module, split sizes, per-class metrics in own protocol. Not a usable baseline until reproduced

						97.8%, 'OCT' 98.7%). Values equal 44/48, 46/48, 47/48 → possibly tiny eval sample (unverified)		finance); single dataset; compute cost; no latency or interpretability analysis. Headline numbers unverified	vel or external validation; no per-class/ macro metrics	
MacularNet: Towards Fully Automated Attention-Based Deep CNN for Macular Disease Classification	Sapna S. Mishra, Bappaditya Mandal, Niladri B. Puhan	2022	Duke, NEH, UCSD/Kermany, OCTID	Image-level macular OCT classification (3-, 4-, 5-class); layer deformation as motivation only	CNN + attention. VGG-Face base, transfer learning (VGG16, ResNet50 tested). 'Deformation-aware' attention = 8-head scaled dot-product self-attention on 14×14 Conv5_3 map; spatial only. No RoI/denoising/flattening/segmentation	UCSD: Acc 92.60±1.48, P 90.27, R 88.52 (LACNN 90.10/86.20/86.80); ~16.56 M weights. Duke: LPO acc 97.45 vs 5-fold CV 99.94. NEH: LPO 94.18 vs CV 99.79. OCTID: Acc 93.12±8.59, AUC 98.51. Attention ablation (one LPO case): Duke 98.50 vs 95.88; NEH 97.38 vs 96.34. No per-class UCSD results	Reports LPO next to k-fold CV (gap 2.5 pts Duke, 5.6 pts NEH); 4 datasets; base-network substitution study; ~3.5× parameter reduction attributed to attention; no RoI preprocessing; attention-map evolution shown	UCSD result from low-data, 112×112 protocol; not comparable to official-split results; patient grouping of UCSD subsets/CV folds unstated (LPO–CV gap suggests leakage); ablation on one LPO case; attention-RPE alignment qualitative only; face-pretrained base; no cross-dataset test; subtler/early-stage disease untested	No explicit retinal-layer relation model (generic spatial self-attention); no quantitative attention-lesion/layer overlap; no per-class UCSD results; no cross-dataset generalization; no graph reasoning	Closest precedent for attention component and anatomy-motivated argument without layer labels. Adopt: LPO vs CV reporting, no RoI pipeline, multi-head self-attention on CNN maps, attention-map evolution. Yours must show graph adds beyond this; test attention alignment quantitatively
Diagnosis of retinal diseases using the vision transformer model based on optical coherence tomography images	Zenan Zhou, Chen Niu, Huanhuan Yu, Jiaqing Zhao, Yuchen Wang, Cuixia Dai	2023	Kaggle kermany2018	4-class image-level OCT classification	Vision Transformer: patch tokens, linear embedding, class token, multi-head self-attention encoder, MLP head; symmetric cross-entropy. Compared with EfficientNet, ResNet50, VGG16 (same hyperparameters). Patch size, depth, pre-training, augmentation not reported	Test: acc 95.76%, sensitivity 95.77%, specificity 98.59% (aggregation unstated). Validation acc 96.57%, loss 0.0929. Val/test acc: EfficientNet 85.35/83.25; ResNet50 88.12/85.39; VGG16 92.85/91.42; ViT 96.57/95.76. Per-class: confusion matrix figure only	Same-protocol comparison vs 3 CNNs; sensitivity and specificity reported; official 968-image test set	'Statistically significant' claimed, no test reported; best of 30 runs kept (optimistic selection); text mentions 5 CNNs, table shows 3; official split overlaps by subject ID (Tampu et al.), unaddressed; attention claim without attention maps; no limitations stated	No class-wise P/R in text; no interpretability or attention maps; no patient-level or external validation; no graph/anatomical modeling	ViT baseline on official Kaggle split; global-attention comparator. Doesn't show what attention attends to → gap your project can address
Harnessing the power of vision transformers for enhanced OCT image classification	Elena-Anca Paraschiv, Alina-Elena Sultana	2024	Kaggle Kermany)	4-class OCT classification	(1) ViT from scratch (dim 64, depth 8, 8 heads, MLP 128; 20 epochs; lr 3e-4; Adam; weighted sampler). (2) DeepViT. (3) Hybrid: ImageNet ResNet50 → linear projection → ViT blocks → FC head (10 epochs, lr 1e-3). Self-attention only; no graph	Test Acc/P/R/F1: ViT 96.80/96.80/96.80/0.96; DeepViT 89.98/90.03/89.98/0.90; Hybrid 99.97/99.10/99.97/0.99. Text/conclusion give 99.38% for hybrid; 99.97% impossible on 968 images (1 error = 99.90%); 99.38% = 6 errors but text reports 8 Drusen errors. Drusen confusions: ViT 7→CNV, 7→Normal; DeepViT 19, 17. Averaging unstated	Official split with class counts; weighted sampler for imbalance; CNN+transformer hybrid; Drusen identified as most confused class; explicit limitations section	Single dataset; heavy compute (DeepViT); hybrid complexity; limited hyperparameter/augmentation search; 32-image validation set; hybrid pre-trained vs ViT from scratch (confounded comparison); numeric inconsistencies; no repeated runs; Table 3 mixes protocols (lists 95.77 sensitivity as accuracy); no attention visualization; possible patient overlap in v2 split, unaddressed	No patient-level/leakage analysis; no interpretability; no per-class tables in text; no external validation; no graph/anatomical modeling	Supports CNN-then-attention (hybrid) design; Drusen hard class → track per-class recall. Fair comparison must control pre-training and report seeds

Inflation of test accuracy due to data leakage in deep learning-based classification of OCT images	Iulian Emil Tampu, Anders Eklund, Neda Haj-Hosseini	2022	AIIMS, Srinivasan, Kermany v2, Kermany v3	Evaluation methodology for 2D-slice OCT classification; effect of original, per-image, per-subject splits; random-label leakage probe	LightOCT (lightweight CNN) retrained per split; 10× repeated 5-fold CV; automated subject-ID overlap check across original splits. No attention/graph model	MCC (primary), AUC, F1, Acc, P, R (mean±SD, 50 models). Kermany v2: original acc 0.911 (MCC 0.886); per-image 0.770 (MCC 0.707); per-subject 0.669 (MCC 0.588). v3: original 0.704; per-image 0.738; per-subject 0.671. Srinivasan MCC 0.853 (per-image) vs 0.426 (per-subject). Inflation: MCC 0.07-0.43 (acc 5-30%). Subject overlap: 92% in v2, none in v3	Quantifies leakage on Kermany; version dependence shown; review table of published OCT splits; lists other leakage sources (test-set tuning, normalization statistics, augmentation)	Only train-test overlap studied (not hyperparameter, normalization, augmentation leakage); datasets distributed pre-augmented, split timing unknown; single small model → absolute accuracies (0.67-0.77) not predictive of strong models; '92% overlap' definition needs checking in methods	No attention/graph models under leakage-free splits; unclear if near-duplicate neighbouring B-scans remain after subject-level split	Defines evaluation protocol: subject-level split, no test-set tuning, normalization fit on train only, check Kermany version. Kaggle layout = v2 → test patient overlap before training
Retinal graph neural network for segmentation of retinal vasculature and foveal avascular zone from OCTA images	Nisan Pranavah Raja, Anju Thomas, Bibin Francis, Varun P. Gopi, Palanisamy Ponnusamy	2026	OCTA-500, ROSE-0	Pixel-level segmentation of vessels and FAZ in OCTA (not classification)	RGNNNet: residual CNN encoder-decoder + Retinal GNN block on bottleneck. Dense learned affinity (1×1 conv projections), symmetrized, degree-normalized, propagates third projection; skip connection; Dice+Focal loss. Dense graph, not kNN; close to non-local self-attention. Pixel-level supervision	FAZ Dice 96.78% (6 mm), 98.02% (3 mm). ROSE-0: Dice 0.915, IoU 0.843, acc 0.994, sens 0.908, spec 0.997 (only after 2-98% percentile stretch; Dice ≈ 0 without). 0.83 M params; 11.25 ms/image. Friedman $\chi^2(11)=63.06$, $p<0.001$; mean rank 1.00. FAZ kappa 0.951 (95% CI 0.941-0.960), 126 scans	Statistical tests (Friedman/ Nemenyi, Wilcoxon, effect sizes); ablations (affinity type, normalization); external test with domain shift reported; affinity visualization; very small model	Different modality and task; no exhaustive hyperparameter search; external test limited to ROSE-0; single-plexus input; no DR staging; 'clinical validation' = agreement with reference masks, not reader study; 126 scans vs 50-image test split unexplained; interpretability qualitative (two queries); only 8 paired blocks → limited test power	No classification or image-level-label setting; unclear if learned affinity matches anatomical structures in OCT B-scans	Shows graph building from CNN feature maps (affinity, symmetrization, degree normalization) and testing (ablations, statistics, affinity visualization). Differs: image-label classification, B-scan structure; must check graph adds beyond self-attention (dense affinity here is attention-like)
Vision GNN: An Image is Worth Graph of Nodes	Kai Han, Yunhe Wang, Jianyuan Guo, Yehui Tang, Enhua Wu	2022	ImageNet-1k	General image recognition and detection backbone	Patches as nodes; kNN edges in feature space. ViG block = Grapher (graph conv, dilated aggregation) + per-node FFN vs over-smoothing. Edge-conv, GIN, GraphSAGE, max-relative tested (max-relative default). Position encoding; isotropic and pyramid versions	ImageNet top-1/top-5: Pyramid ViG-Ti 78.2/94.2 (10.7 M, 1.7 GFLOPs); ViG-S 82.1/96.0 (27.3 M, 4.6 GFLOPs); ViG-M 83.1/96.4; ViG-B 83.7/96.5. Detection results reported (not extracted)	Patch-graph design reused by OCT papers; ablations (graph conv type, FFN); deeper layers link semantically related nodes (qualitative)	No limitations section; no medical data; edges follow appearance, not anatomy; neighbour selection analysed qualitatively only; per-layer kNN construction cost not analysed	No medical images, class imbalance, patient-level evaluation, or lesion-level interpretability	Template for GCN stage (kNN nodes, FFN, positional encoding). Design choice: nodes from raw patches (ViG) vs CNN feature-map locations (PVGCN, RGNNNet); edges from feature similarity vs spatial/layer prior